## Defence Tech, Security & Disaster Management

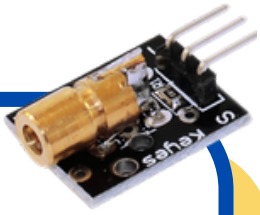


# Project Name: Laser Security System



**The Problem:** Aarav is helping to protect a room where important equipment is kept. The security guard cannot watch the entrance all the time, especially at night. Aarav worries that someone may enter the room without permission. He decides to build a smart system that can detect when someone crosses the entrance.

🎯 **What You Will Design:** Students will build a smart defense protection system that detects unauthorized entry using a laser security barrier. Authorized users can access the area using an RFID card, while intruders trigger an alarm and a warning message.



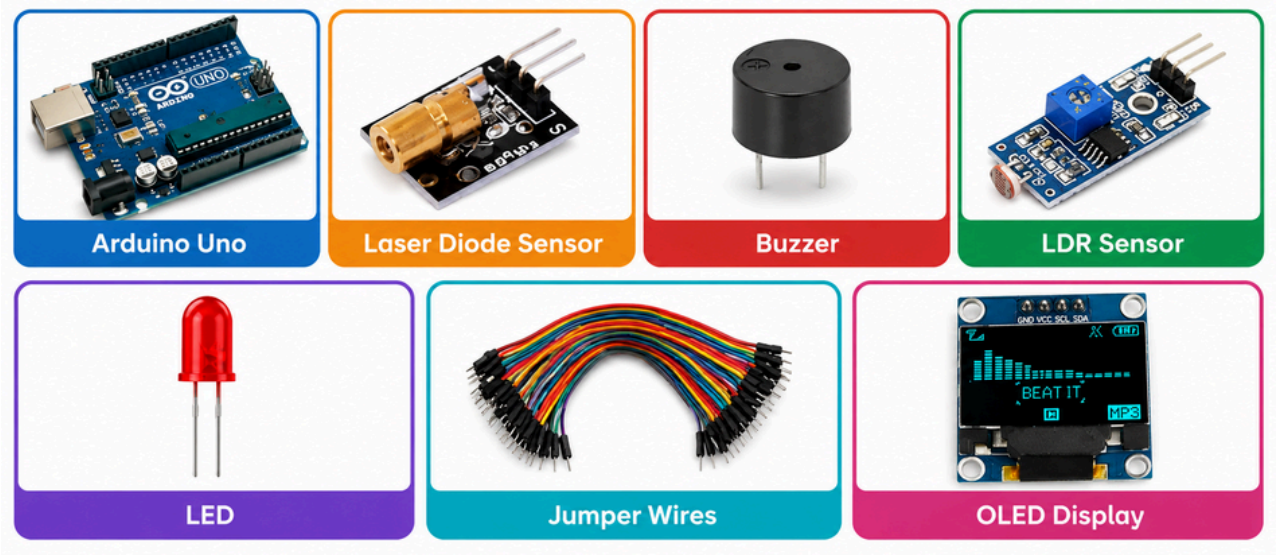
### 🔴 What is a Laser Diode Sensor?



A laser diode sensor uses a laser beam to create an invisible security line. When someone crosses or blocks the laser beam, the system can detect the interruption.

**Example:** 🏦 Similar laser security systems are used in banks, museums, and high-security buildings to detect unauthorized entry.

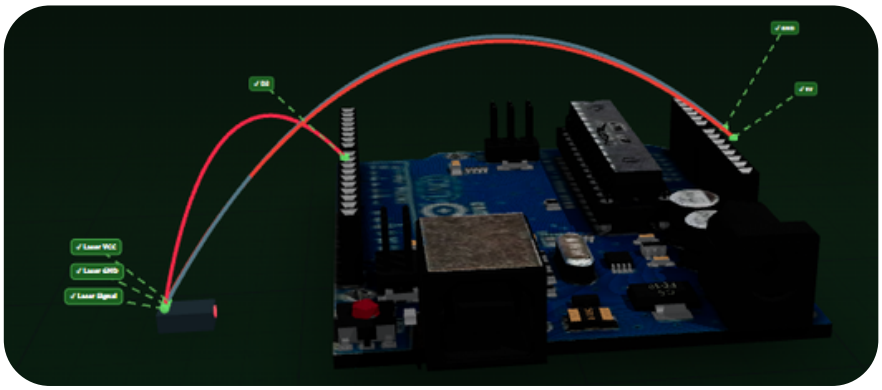
## Components Required



## Circuit Designing

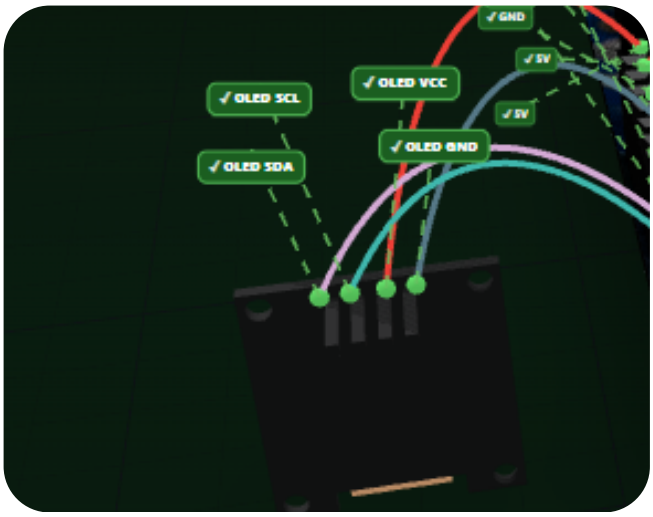
**Step 1: Connect laser diode to the Arduino Uno.**

| Laser Pins | Arduino Pins |
|---|---|
| Vcc Pins | 5V Pin |
| GND Pin | GND Pin |
| Signal Pin | Digital Pin 2 |



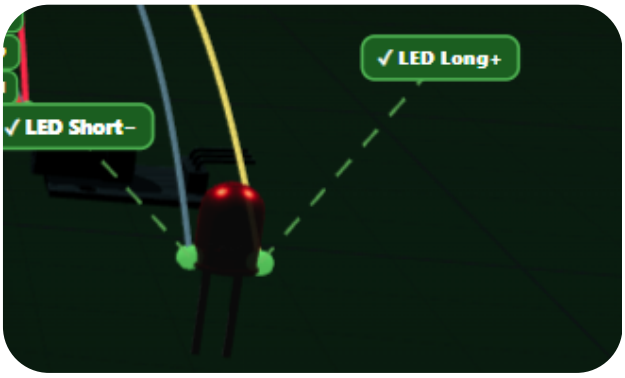
**2. Connect the OLED display to the Arduino Uno.**

| OLED Pins | Arduino Pins |
|---|---|
| Vcc Pins | 5V Pin |
| GND Pin | GND Pin |
| SDA Pin | Analog A4 Pin |
| SCL Pin | Analog A5 Pin |



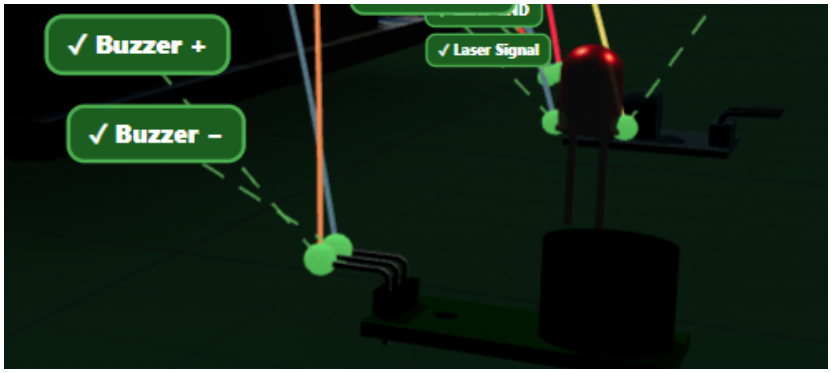

**3. Connect the LED to the Arduino Uno.**

| LED Pins | Arduino Pins |
|---|---|
| Long Leg | Digital Pin 7 |
| GND Pin | GND Pin |



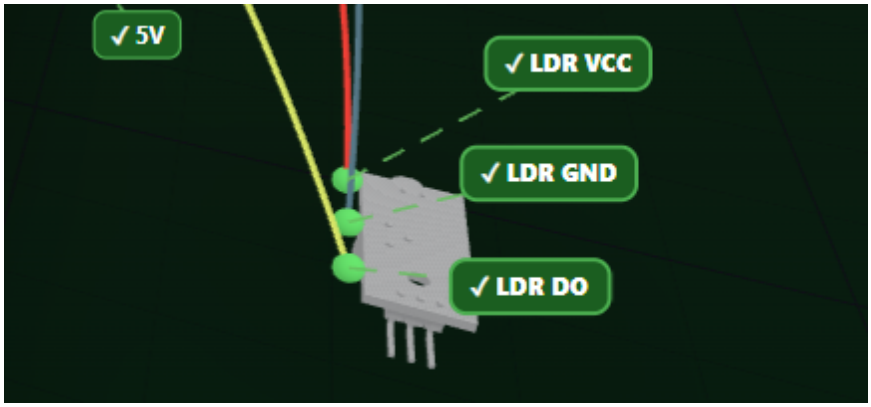
### 4. Connect the buzzer to the Arduino Uno

| Buzzer Pins | Arduino Pins |
|---|---|
| Positive Pin | Digital Pin 8 |
| Negative Pin | GND Pin |




### 5. Connect the LDR sensor to the Arduino Uno,

| LDR Pins | Arduino Pins |
|---|---|
| DO pins | Digital Pin 9 |
| Vcc pIn | 5V Pin |
| GND Pin | GND |




## Project Coding

### Step 1: Include Libraries and Define Component Pins

```
#include <Wire.h>
#include <Adafruit_GFX.h>
#include <Adafruit_SSD1306.h>

const byte LASER_PIN  = 2;
const byte LDR_PIN    = 9;
const byte LED_PIN    = 7;
const byte BUZZER_PIN = 8;

// Set according to the buzzer type
const bool ACTIVE_BUZZER = false;

const unsigned long FILTER_MS = 30;
```

### Step 2: Initialize the OLED Display and Program Variables

```
Adafruit_SSD1306 oled(128, 64, &Wire, -1);

bool displayReady = false;
bool alarmActive = false;

int beamIntactLevel;
int previousReading;

unsigned long lastChange;

bool addressFound(byte address) {
  Wire.beginTransmission(address);
  return Wire.endTransmission() == 0;
}
```

### Step 3: Create the Alarm Control Function

```
void updateAlarm(bool enabled) {
  alarmActive = enabled;

  digitalWrite(LED_PIN, enabled ? HIGH : LOW);

  if (ACTIVE_BUZZER) {
    digitalWrite(BUZZER_PIN,
                 enabled ? HIGH : LOW);
  } else {
    if (enabled) {
      tone(BUZZER_PIN, 2000);
    } else {
      noTone(BUZZER_PIN);
      digitalWrite(BUZZER_PIN, LOW);
    }
  }

  Serial.println(enabled ? F("ALERT: Beam interrupted")
                         : F("Beam intact"));
```

### Step 4: Display the Security Status on the OLED

```
if (!displayReady) return;

oled.clearDisplay();
oled.setTextSize(1);
oled.setTextColor(SSD1306_WHITE);

oled.setCursor(0, 0);
oled.println(F("LASER SECURITY"));
```

### Step 5: Make code to print the information on the OLED Display

```
  if (enabled) {
    oled.println(F("BEAM INTERRUPTED"));
    oled.println(F("ALARM ACTIVE"));
  } else {
    oled.println(F("Beam intact"));
    oled.println(F("Monitoring..."));
  }

  oled.display();
}
```

### Step 5: Configure the Components in setup()

```
void setup() {
  Serial.begin(9600);

  pinMode(LASER_PIN, OUTPUT);
  pinMode(LDR_PIN, INPUT);
  pinMode(LED_PIN, OUTPUT);
  pinMode(BUZZER_PIN, OUTPUT);

  digitalWrite(LED_PIN, LOW);
  digitalWrite(BUZZER_PIN, LOW);
  digitalWrite(LASER_PIN, HIGH);

  Wire.begin();
```

```
  byte address = 0;

  if (addressFound(0x3C)) {
    address = 0x3C;
  } else if (addressFound(0x3D)) {
    address = 0x3D;
  }

  if (address != 0) {
    displayReady =
        oled.begin(SSD1306_SWITCHCAPVCC, address);
  }
```

### Step 6: Calibrate the Laser Beam and Sensor

```
if (displayReady) {
  oled.clearDisplay();
  oled.setTextSize(1);
  oled.setTextColor(SSD1306_WHITE);
  oled.setCursor(0, 0);

  oled.println(F("CALIBRATING"));
  oled.println(F("Aim laser at LDR."));
  oled.println(F("Keep beam clear."));
  oled.display();
} else {
  Serial.println(F("OLED unavailable."));
}
```

```
  delay(3000);

  int candidate = digitalRead(LDR_PIN);
  unsigned long stableSince = millis();

  while (millis() - stableSince < 1000UL) {
    int reading = digitalRead(LDR_PIN);

    if (reading != candidate) {
      candidate = reading;
      stableSince = millis();
    }

    delay(5);
  }
```

```
  beamIntactLevel = candidate;
  previousReading = candidate;
  lastChange = millis();

  updateAlarm(false);
}
```

**Step 7: Detect Beam Interruption and Filter Brief Changes**

```
void loop() {
  int reading = digitalRead(LDR_PIN);
  unsigned long now = millis();

  if (reading != previousReading) {
    previousReading = reading;
    lastChange = now;
  }

  if (now - lastChange >= FILTER_MS) {
    bool interrupted =
        (reading != beamIntactLevel);

    if (interrupted != alarmActive) {
      updateAlarm(interrupted);
    }
  }
}
```



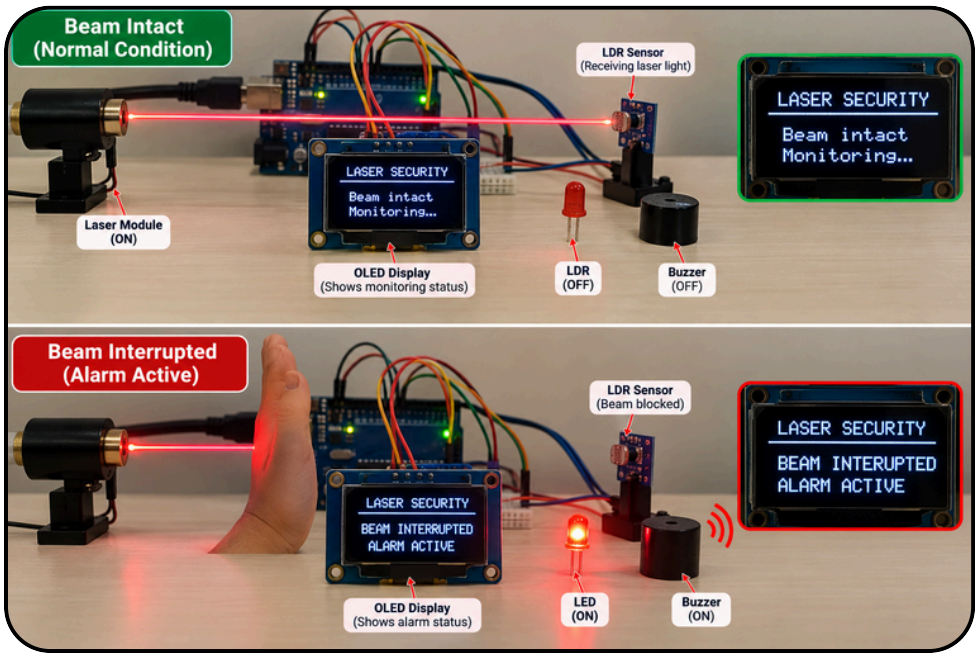
## 🌾 Where Is This Used in Real Life?

Similar security systems are used in military facilities, warehouses, museums, and restricted buildings to detect unauthorized movement. They provide quick alerts to security teams and help protect important areas.

## Learning Outcomes

- The students will be able to use a laser sensor to detect movement in a protected area.
- The students will be able to write code that activates alerts based on sensor inputs.
- The students will be able to build an automatic security system using sensors.

## Project Output






## Section 1: Multiple Choice Questions

**1. What problem does Aarav want to solve?**

a) The room is too small
b) Someone may enter without permission
c) The lights are too bright
d) The door is too heavy

**2. Which component detects when the security laser beam is blocked?**

a) OLED Display
b) Buzzer
c) Laser Diode Sensor
d) Jumper Wires

**3. What happens when an intruder is detected?**

a) The buzzer and LED turn on
b) The Arduino turns off
c) The laser becomes brighter
d) The wires disconnect

## Section 2: True and False Statements

**1. A laser beam can be used to create an invisible security line.**

.....

**2. The buzzer produces an alert sound when an intruder is detected.**

.....

**3. The OLED display is used to detect movement.**

.....